
Your non-physics major friend saw on YouTube that matter and antimatter could annihilate to produce photons. This friend then comes to you claiming to have discovered something profound: “The inverse process of creating matter plus antimatter from a single photon such as

$$\gamma \rightarrow e^+ + e^- \quad (1)$$

must occur!” Convince your friend who only learns physics from YouTube (instead of learning from yourself™) that this process can not occur

Proof. For a photon, it has four momentum

$$p_\gamma^\mu p_{\gamma\mu} = p_\gamma^2 = 0 \quad (2)$$

Four momentum conservation requires the positron and electron momenta to equal the photon

$$p_\gamma = p_- + p_+$$

Hence

$$0 = p_\gamma^2 = (p_- + p_+)^2$$

But, since $p_-^2 = p_+^2 = m_e^2 c^2 > 0$, then

$$\begin{aligned} (p_- + p_+)^2 &= p_-^2 + p_+^2 + 2p_- \cdot p_+ \\ &= 2m_e^2 c^2 + 2p_- \cdot p_+ \\ &> 0 \end{aligned}$$

A contradiction. Therefore

$$\gamma \not\rightarrow e^- + e^+ \quad (3)$$

Pair production requires another particle such as a second photon, a proton, or a nucleus to conserve four-momentum $\square$